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Module 2
Opening to Depressive and Bipolar Disorders

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Module 6
Treatments for Depressive and Bipolar Disorders

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- ✓ Overview: Biological Treatments
  Reading • 5 min
- ○ Medications and Neurotransmitters
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Lesson 10: Psychological Treatments
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Medications and Neurotransmitters

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Antidepressant Medications

Antidepressant medications, which are used to alleviate the symptoms of depression, were initially developed in the 1950s. The earliest antidepressant medications included two groups, monoamine oxidase inhibitors (MAOIs) and tricyclic antidepressants (TCAs).

Monoamine Oxidase Inhibitors & Tricyclic Antidepressants

Monoamine oxidase inhibitors (MAOIs) are a group of antidepressant medications that function in presynaptic neurons by inhibiting the activity of the enzyme monoamine oxidase, which typically works to break down monoamines, including serotonin, norepinephrine and dopamine, and make them inactive. By preventing the inactivation of these neurotransmitters, MAOIs allow more of them to remain available in the brain (in presynaptic neurons and in synapses for binding to postsynaptic receptors), which helps to alleviate the symptoms of depression.

MAOIs are now prescribed less often, however, in part because of their side effects. The major side effect of MAOIs is that they can interact with prescription and nonprescription medications and some foods to cause dangerous conditions such as a "hypertensive crisis," which is when your blood pressure spikes up and can lead to a stroke, heart attack, or other life threatening problems. To avoid this side effect, people on MAOs must maintain strict diets that require them to avoid a large number of foods (Yamada & Yasuhara, 2004).

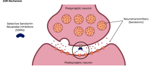
Tricyclic antidepressants are the other groups of medications developed in the 1950s as a treatment for depression, and are still prescribed today. **Tricyclic antidepressants (TCAs)** are medications that act by blocking the reuptake of serotonin, dopamine, and/or norepinephrine into the presynaptic neuron, thereby increasing the amount of neurotransmitter available in the synapse for binding to postsynaptic receptors.

Tricyclics are generally prescribed less often than in the past, as they can have a variety of side effects, including dry mouth, blurred vision, constipation, and urinary retention. In addition, overdose of TCAs can lead to sudden cardiac arrest, which is another reason they are not prescribed as widely as they were in the past (Santarsieri & Schwartz, 2015).

More Recent Antidepressant Medications

Since the 1950s, other groups of antidepressant medications, including selective serotonin reuptake inhibitors (SSRIs) and serotonin-norepinephrine reuptake inhibitors (SNRIs) have increased in use due to having more targeted physiological mechanisms (e.g., target specific neurotransmitters) and fewer side effects. **Selective serotonin reuptake inhibitors (SSRIs)** are medications that work by blocking the reuptake (e.g., reabsorption) of the neurotransmitter serotonin into serotonin-containing presynaptic neurons, thus increasing the amount of serotonin in the synapse; higher levels of serotonin in the synapse are correlated with elevated mood. SSRIs are the most commonly prescribed antidepressants in the United States (64.2% in 2015; Luo et al., 2020) and many other countries (e.g., Luo et al., 2020; Yu et al., 2020). In fact, you may be familiar with the brand names of many SSRIs, which are often widely advertised, including Zoloft (Sertraline), Prozac (Fluoxetine), Celexa (Citalopram), Lexapro (Escitalopram), and Paxil (Paroxetine) (FDA, 2019). Common side effects of SSRIs include sexual dysfunction, sleepiness, and weight gain. Yet, only 25% of those who experienced one or more side effects considered them to be "very bothersome" or "extremely bothersome" (Cascade et al., 2009). **Serotonin-norepinephrine reuptake inhibitors (SNRIs)** work in the same manner as SSRIs but target norepinephrine as well as serotonin. These "dual-action" antidepressants have similar side effects as SSRIs, and have a slight higher rate of remission for depression (Sussman, 2003). The *SSRI Mechanism* figure illustrates the general process for blocking reuptake for these antidepressants, which increases the amount of neurotransmitters available in the synapse.

SSRI Mechanism


Presynaptic neuron
Neurotransmitters (Serotonin)
Selective Serotonin Reuptake Inhibitors (SSRIs)
Postsynaptic neuron

Note. The presynaptic neuron contains vesicles filled with neurotransmitters that are released into the synapse, or the space between the presynaptic and postsynaptic neurons. Some of the neurotransmitters bind with receptors on the postsynaptic neuron and others are recycled through reuptake back into the presynaptic neuron. The presence of psychotropic drugs, such as SSRIs, on the presynaptic neuron prevents (blocks) the reuptake process.

Ketamine is a drug that is closely related to phencyclidine (PCP) that was developed in the 1960s as a dissociative anesthetic for humans and nonhuman animals. Since then it has been used as a party drug, and researchers have discovered its promising use as an antidepressant medication. In 2000, Robert Berman and colleagues published the results of a clinical trial which documented the efficacy of ketamine as an antidepressant. Continued research has demonstrated the unique benefits of ketamine, including that it is fast-acting (in up to 3 hours) and is long-lasting (up to 7 days) (Zanos & Gould, 2018). These benefits are particularly important when treating individuals with suicidal ideation. The drawbacks of ketamine, however, have limited its use. For example, side effects include dissociative effects and changes in sensory perception. Other drawbacks include the intravenous route of administration and the potential for abuse (Zanos & Gould, 2018). For these reasons, ketamine is often used for cases in which individuals experience treatment-resistant depression. While the exact molecular mechanisms of the antidepressant effects of ketamine are not known, continued research suggests that through one or multiple paths, ketamine increases the amount of excitatory neurotransmitters available in the brain, which is a key component of decreasing the symptoms of depression (Jelen & Stone, 2020; Zanos & Gould, 2018). For instance, by blocking N-methyl-D-aspartate (NMDA) receptors, ketamine indirectly increases glutamate release, which has been indicated in decreasing the symptoms of depression (Jelen & Stone, 2020).

The Placebo Effect and Depression

You may have heard the term **placebo** in reference to a medical or psychological intervention or treatment that is believed to be "inert." This makes it valuable as a control condition against which to compare an intervention or treatment of interest. The **placebo effect** then is a clinically significant response to a placebo, deriving from the recipient's expectations or beliefs regarding the intervention. A number of meta-analyses and systematic reviews have concluded that much of the effectiveness of antidepressant use for patients with mild to moderate depression is due to a placebo effect in which patients' expectancy that the medication will be effective is responsible for the changes in the patients' depressive symptoms (see Fournier et al., 2010; Kirsch, 2019). While not everyone agrees that the effectiveness of antidepressant medications is solely due to a placebo effect, the evidence is relatively strong, especially for mild and moderate levels of depression. There is also consensus that for severe cases of depression, the pharmacological effects of antidepressants are effective in treating depressive symptoms (Fournier et al., 2010).

Bipolar Medications

Ideally, treatments for bipolar disorder target chronic mood instability, including both manic and depressive episodes. Often, however, acute treatments for either manic or depressive episodes are provided instead of addressing the fluctuations in mood (Geddes & Miklowitz, 2013). For example, antidepressant medications such as SSRIs and SNRIs are commonly prescribed to patients with bipolar disorder (Poranen et al., 2022). This use of antidepressants continues to be debated, as there is little evidence for the long-term effectiveness of this treatment strategy (Kato, 2019). In terms of mania, antipsychotics are often used to target these acute episodes, but they are also not likely to be effective as a long-term mood stabilizing treatment (Kato, 2019).

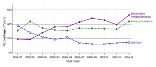
Lithium & Mood Stabilizing Medications

One of the earliest treatments for bipolar disorder, lithium, remains the most effective for long term treatment, including both manic and depressive episodes (Alda, 2015). Lithium is known to affect multiple signaling pathways, and other cellular processes, but the exact mechanisms through which it prevents manic and depressive episodes remains unclear. It is also known to have variability in its effectiveness–while some patients have a full response to symptoms of bipolar disorder, others show only a partial response, or no response to the medication (Lin et al., 2020). In addition, lithium can have serious side effects, such as weight gain, cognitive impairment, and negative effects on organs including the kidneys, thyroid gland, and parathyroid glands, among others (Gitlin, 2016; Lin et al., 2020). Given these concerns, the use of lithium for bipolar disorder has declined over time in a number of countries, including the United States, Finland, Hong Kong, and the United Kingdom (Lin et al., 2020; Ng et al., 2021; Poranen et al., 2022). Given lithium's potential for robust treatment, however, including lower rates of relapse, suicide prevention, and lower rates of hospitalization, many have encouraged more frequent use of this medication as a primary treatment for bipolar disorder (Ng et al., 2022; Poranen et al., 2022).

Other mood stabilizing medications have increased in use for the treatment of bipolar disorder over time, including anticonvulsants and atypical antipsychotics (Lin et al., 2020; Ng et al., 2022; Poranen et al., 2022). Anticonvulsants, while conventionally used to treat conditions such as epilepsy, have been useful in stabilizing mood for patients with bipolar disorder. It is not clear if the mood stabilizing mechanisms are the same as the anticonvulsant mechanisms. Atypical antipsychotics also function to decrease both manic and depressive episodes (Kato, 2019).

The U.S. trends from 1996 to 2015 for lithium, anticonvulsants, and secondary antipsychotics (e.g., atypical antipsychotics) are illustrated in the *Mood Stabilizer Trends* graph (Lin et al., 2020).

Mood Stabilizer Trends



Note. A line graph with the percentage of visits among patients diagnosed with bipolar disorder on the y-axis, ranging from 0% to 60%, and visit year on the x-axis, ranging from 1996-1997 through 2014-2015. Three lines are used to depict the three medications: secondary antipsychotics, anticonvulsants, and lithium. Secondary antipsychotics began near 20% at 1996-1997 and then increased over time to over 50% reported at 2014-2015. Anticonvulsants started higher, near 30%, at 1996-1997 and then increased a smaller amount to just over 40% at 2014-2015. Lithium began the highest of the three categories, at nearly 40% in 1996-1997, and decreased over time to under 20% in 2014-2015.

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